CAP 931: SalesIntel AI

Building an AI Sales Assistant

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CAP 931 Capstone Project

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Overview

For my CAP 931 project, I built SalesIntel AI. It helps a salesperson research a company before a meeting. The user enters product and company details. The app collects public information, checks the AI findings against the sources, and creates a one-page sales brief.

I used Python, uv, Streamlit, Groq, Requests, BeautifulSoup, Pydantic, pypdf, ReportLab, pytest, and Ruff. Groq runs three research roles: company strategy, competitor research, and leadership research. These are focused model calls managed by Python, not independent agents.

The app is a local student project. It reports missing evidence instead of guessing, but a person should still review the results.

Steps

Step 1. Set up the project

I created a Python 3.12 project with uv. I kept the application code in `src/cap_931` and separated the interface, prompts, research, document upload, PDF generation, and tests.

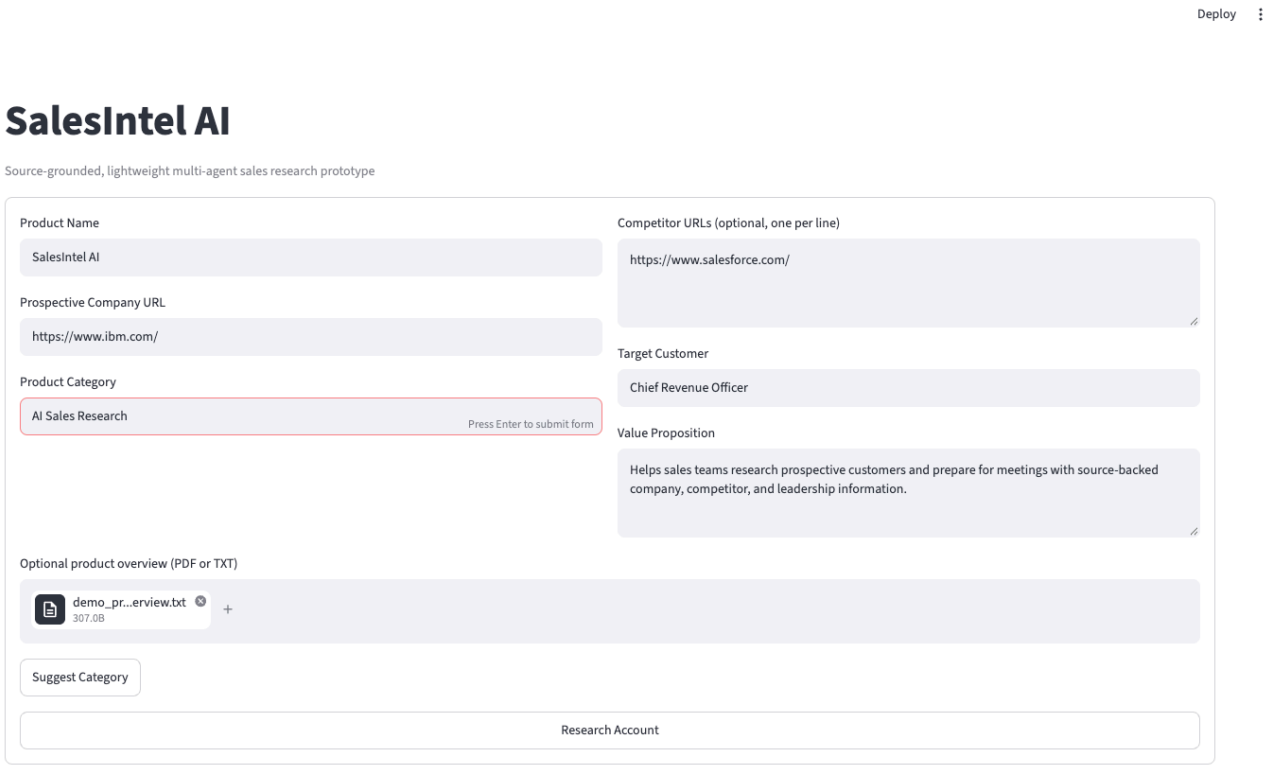
CAP 931 Project Structure

```
CAP_931/
  README.md
  artifacts/demo_product_overview.txt
  artifacts/experiments/final_prompt_comparison.json
  artifacts/experiments/final_prompt_manual_scores.json
  artifacts/experiments/manual_scores.json
  artifacts/experiments/model_runs.json
  artifacts/experiments/source_snapshots.json
  docs/CAP931_Zaheera_KB_Report.pdf
  docs/deployment_plan.md
  docs/evidence/final_test_results.txt
  docs/evidence/representative_research_run.txt
  docs/experiments.md
  docs/report.md
  docs/sales_intelligence_brief.pdf
  docs/screenshots/01-project-structure.png
  docs/screenshots/02-streamlit-inputs.png
  docs/screenshots/03-input-validation.png
  docs/screenshots/04-source-details.png
  docs/screenshots/04-source-discovery.png
  docs/screenshots/05-agent-results-continued.png
  docs/screenshots/05-agent-results.png
  docs/screenshots/06-product-upload.png
  docs/screenshots/07-pdf-download.png
  docs/screenshots/08-opened-one-page-report.png
  docs/screenshots/09-tests.png
  docs/screenshots/10-prompt-experiment-final.png
  docs/screenshots/11-category-suggestion.png
  docs/screenshots/12-category-accepted.png
  docs/technical_architecture.md
  outputs/prompt_comparison.html
  pyproject.toml
  scripts/capture_static_evidence.py
  scripts/capture_streamlit_evidence.py
  scripts/generate_representative_report.py
  scripts/generate_submission_pdf.py
  src/cap_931/__init__.py
  src/cap_931/agents.py
  src/cap_931/app.py
  src/cap_931/category_suggester.py
  src/cap_931/document_parser.py
  src/cap_931/experiment_runner.py
  src/cap_931/experiments.py
  src/cap_931/llm_client.py
  src/cap_931/prompts.py
  src/cap_931/report_generator.py
  src/cap_931/sales_tools.py
  src/cap_931/schemas.py
  src/cap_931/tests/test_agents.py
  src/cap_931/tests/test_app.py
  src/cap_931/tests/test_category_suggester.py
  src/cap_931/tests/test_discovery.py
  src/cap_931/tests/test_document_parser.py
  src/cap_931/tests/test_experiments.py
  src/cap_931/tests/test_llm_client.py
  src/cap_931/tests/test_sales_tools.py
  src/cap_931/tests/test_step5.py
  src/cap_931/tests/test_validators.py
  src/cap_931/tests/test_web_research.py
  src/cap_931/validators.py
  src/cap_931/web_research.py
  uv.lock
```

Step 2. Build the Streamlit form

I added fields for product name, company URL, product category, competitor URLs, target customer, and value proposition. Competitor URLs are optional. I also added a PDF or TXT upload and a button that can suggest a category.

The final screenshots use SalesIntel AI, IBM, Salesforce, the AI Sales Research category, and a Chief Revenue Officer target.



Step 3. Validate the input

The app checks required fields and blocks unsafe URLs. It rejects empty, oversized, encrypted, broken, and unsupported uploads. Uploaded product text is seller information. It is not evidence about the company being researched.

Deploy

SalesIntel AI

Source-grounded, lightweight multi-agent sales research prototype

Product Name

Cloud Data Platform

Prospective Company URL

https://example.com

Product Category

Data Analytics

Competitor URLs (optional, one per line)

Target Customer

Chief Technology Officer

Value Proposition

Optional product overview (PDF or TXT)

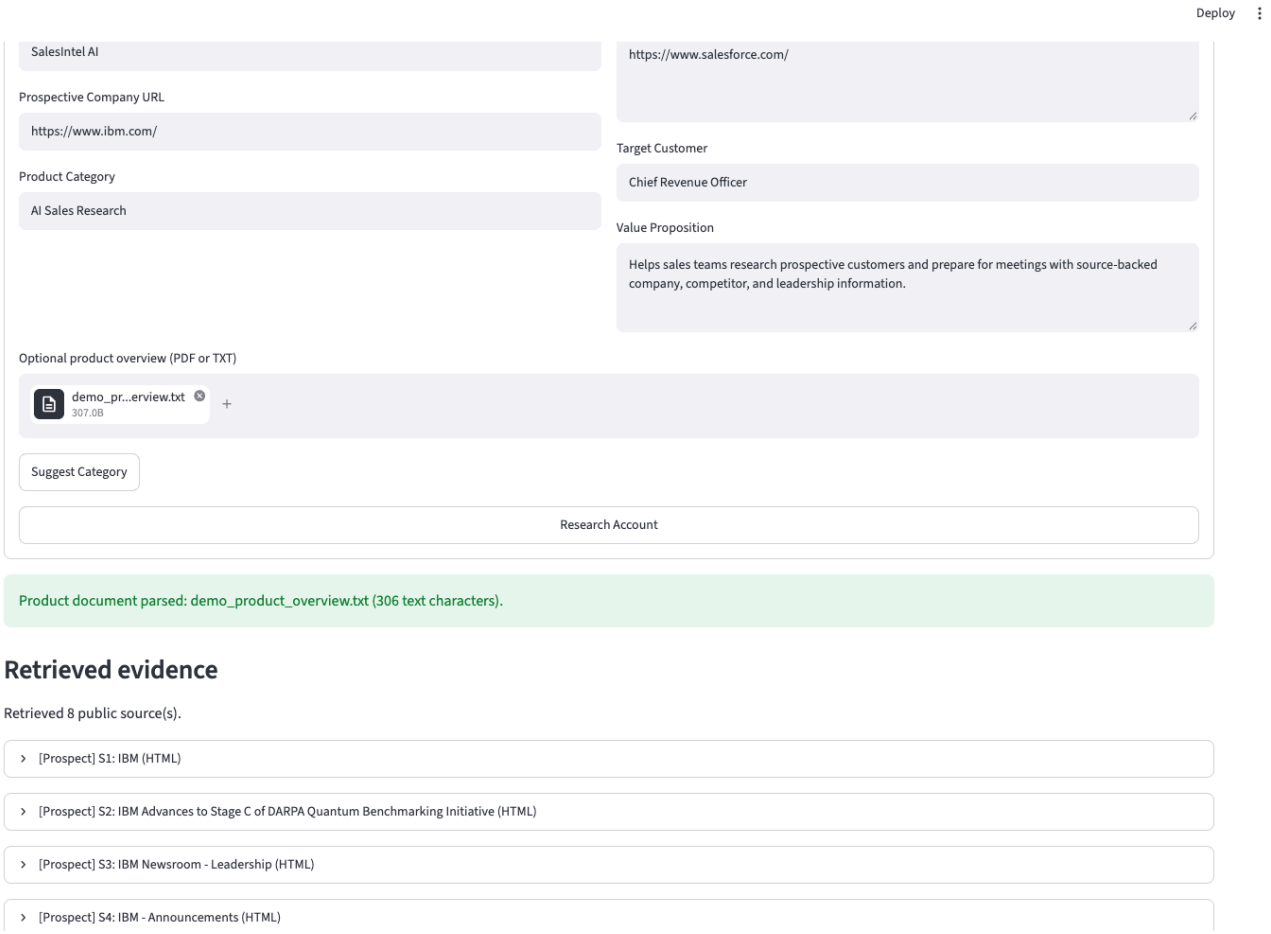
Upload 200MB per file • PDF, TXT

Suggest Category

Research Account

Product Name is required.

Product Category is required.



Step 4. Connect Groq

I first tried Google AI Studio, but I had project permission and subscription errors. I switched to Groq and used the `openai/gpt-oss-20b` model. Groq worked with the Python SDK and could return the structured JSON used by the app.

I wrote separate prompts for company strategy, competitor research, and leadership. Pydantic checks that each response has the expected fields. The Groq free tier also has request and token limits, so the app keeps its source text and model responses bounded.

Step 5. Retrieve public information

The app follows a small number of useful links on company websites. It looks for news, strategy, leadership, investor, product, career, and technology pages. Each successful source receives an ID and a prospect or competitor label.

In the final example, the app retrieved four IBM pages and four Salesforce pages.

Retrieved 8 public source(s).

- [Prospect] S1: IBM (HTML)
- [Prospect] S2: IBM Advances to Stage C of DARPA Quantum Benchmarking Initiative (HTML)
- [Prospect] S3: IBM Newsroom - Leadership (HTML)
- [Prospect] S4: IBM - Announcements (HTML)
- [Competitor] S5: Salesforce: The #1 AI CRM (HTML)
- [Competitor] S6: Salesforce Delivers Record Third Quarter Fiscal 2026 Results Driven by Agentforce & Data 360 - Salesforce (HTML)
- [Competitor] S7: News Explorer - Salesforce (HTML)
- [Competitor] S8: $1 Billion Investment in Spain to Accelerate AI Leadership - Salesforce (HTML)

Agent research results

Verified findings passed source-ID, provenance, and exact-excerpt checks. Sales relevance is a suggestion, not a verified company fact.

Company research

Verified fact: IBM and Siemens are collaborating on research to advance IBM's nanostack architecture

Supporting excerpt (S4): "IBM and Siemens Collaborate on Research to Advance Nanostack Architecture"

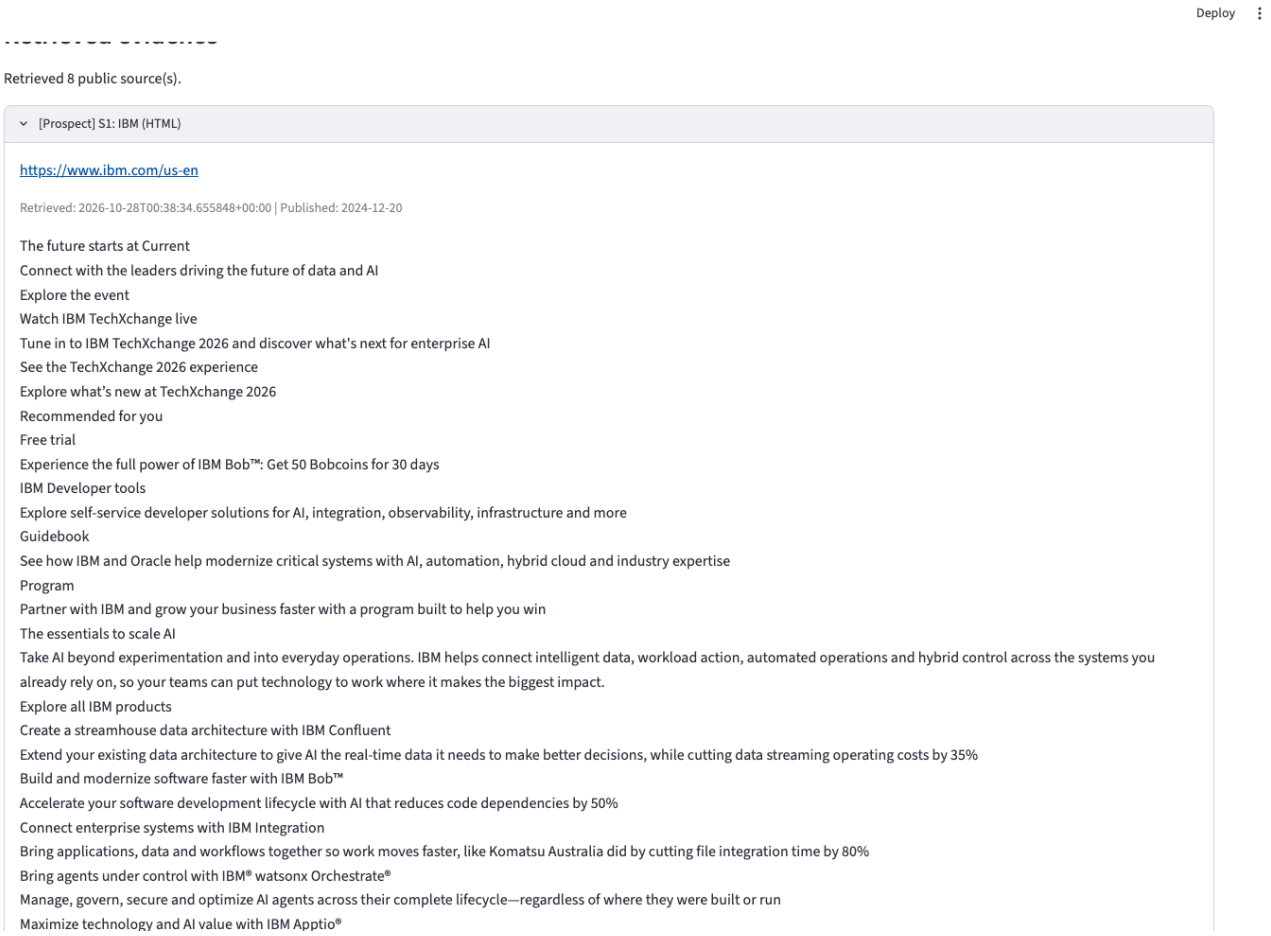
Possible sales relevance: Opportunity to highlight IBM's partnership activities in high-tech research

Source IDs: S4

☑ Include this verified finding in the PDF

Verified fact: IBM and NASA released an open-source AI model to support lunar exploration

Supporting excerpt (S4): "IBM and NASA Release Open-Source AI Model to Support Lunar Exploration"



Step 6. Handle website problems

I added limits for redirects, page size, time, and total research. Redirect loops, blocked pages, and unreadable pages become warnings instead of evidence. The app does not bypass CAPTCHAs, login pages, paywalls, or robots rules.

Microsoft and Oracle blocked some requests during testing. This taught me that public websites are not always available to automated tools.

Step 7. Check the AI findings

Every verified finding needs a source ID and a short excerpt. Python checks that the excerpt appears in the selected source. Sales relevance is shown separately because it is a suggestion, not a verified company fact.

A matching excerpt is helpful, but it does not prove that the complete claim is correct. Important findings still need human review.

Deploy

Verified findings passed source-ID, provenance, and exact-excerpt checks. Sales relevance is a suggestion, not a verified company fact.

Company research

Verified fact: IBM and Siemens are collaborating on research to advance IBM's nanostack architecture

Supporting excerpt (S4): "IBM and Siemens Collaborate on Research to Advance Nanostack Architecture"

Possible sales relevance: Opportunity to highlight IBM's partnership activities in high-tech research

Source IDs: S4

☑ Include this verified finding in the PDF

Verified fact: IBM and NASA released an open-source AI model to support lunar exploration

Supporting excerpt (S4): "IBM and NASA Release Open-Source AI Model to Support Lunar Exploration"

Possible sales relevance: Showcase IBM's involvement in AI for space exploration, relevant to research needs

Source IDs: S4

☑ Include this verified finding in the PDF

Evidence gaps: No explicit mention of IBM's current sales strategy or revenue targets | No data on IBM's procurement plans for AI research tools | No evidence of IBM's internal priorities for sales enablement | No public statements about IBM's competitive positioning against Salesforce | No details on IBM's future product roadmap beyond the cited initiatives

Competitor research

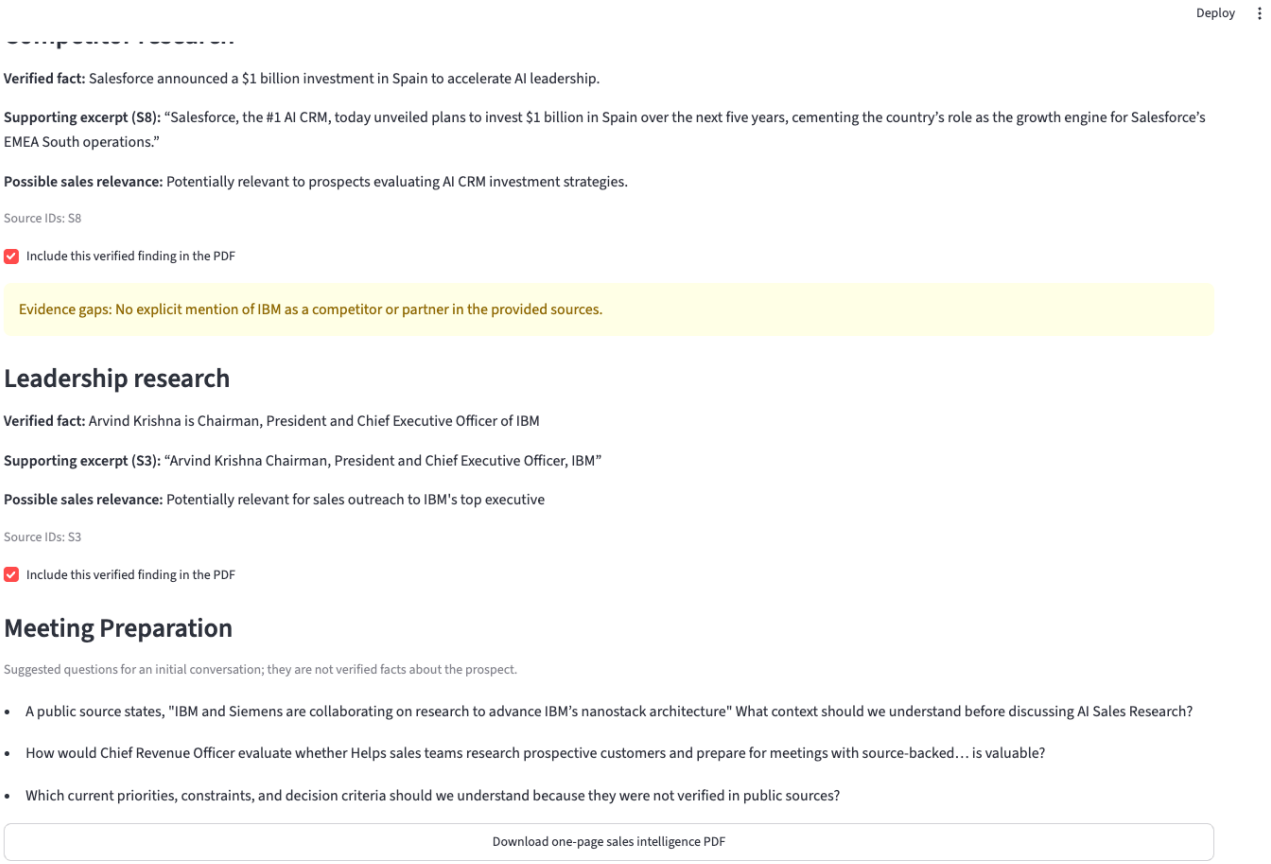
Verified fact: Salesforce announced a $1 billion investment in Spain to accelerate AI leadership.

Supporting excerpt (S8): "Salesforce, the #1 AI CRM, today unveiled plans to invest $1 billion in Spain over the next five years, cementing the country's role as the growth engine for Salesforce's EMEA South operations."

Possible sales relevance: Potentially relevant to prospects evaluating AI CRM investment strategies.

Source IDs: S8

☑ Include this verified finding in the PDF



Step 8. Keep sources separate

Company and leadership findings must use prospect sources. Competitor research can use both source types, but competitor information cannot be reported as a fact about the prospect.

The user can exclude a supported finding before creating the PDF. This review step cannot turn unsupported text into a verified fact.

Step 9. Add meeting questions

The app creates up to three suggested questions from the product details, verified findings, and missing evidence. When little evidence is available, the questions stay neutral instead of assuming that the company has a need.

Step 10. Create the one-page PDF

I used ReportLab to create the sales brief. It includes company, competitor, and leadership sections, meeting questions, evidence gaps, and source links. pypdf checks that the result has exactly one page.

EMEA South operations."

Possible sales relevance: Potentially relevant to prospects evaluating AI CRM investment strategies.

Source IDs: S8

Include this verified finding in the PDF

Evidence gaps: No explicit mention of IBM as a competitor or partner in the provided sources.

Leadership research

Verified fact: Arvind Krishna is Chairman, President and Chief Executive Officer of IBM

Supporting excerpt (S3): "Arvind Krishna Chairman, President and Chief Executive Officer, IBM"

Possible sales relevance: Potentially relevant for sales outreach to IBM's top executive

Source IDs: S3

Include this verified finding in the PDF

Meeting Preparation

Suggested questions for an initial conversation; they are not verified facts about the prospect.

- A public source states, "IBM and Siemens are collaborating on research to advance IBM's nanostack architecture" What context should we understand before discussing AI Sales Research?
- How would Chief Revenue Officer evaluate whether Helps sales teams research prospective customers and prepare for meetings with source-backed… is valuable?
- Which current priorities, constraints, and decision criteria should we understand because they were not verified in public sources?

Download one-page sales intelligence PDF

SALES INTELLIGENCE BRIEF

Prospect: https://www.ibm.com/

Product: SalesIntel AI

Category: AI Sales Research

Target: Chief Revenue Officer

Company Strategy

Verified: IBM and Siemens are collaborating on research to advance IBM's nanostack architecture [S4]

Evidence: "IBM and Siemens Collaborate on Research to Advance Nanostack Architecture"

Possible sales relevance: Opportunity to highlight IBM's partnership activities in high-tech research

Verified: IBM and NASA released an open-source AI model to support lunar exploration [S4]

Evidence: "IBM and NASA Release Open-Source AI Model to Support Lunar Exploration"

Possible sales relevance: Showcase IBM's involvement in AI for space exploration, relevant to research needs

Evidence gap: No explicit mention of IBM's current sales strategy or revenue targets

Competitor Intelligence

Verified: Salesforce announced a $1 billion investment in Spain to accelerate AI leadership. [S8]

Evidence: "Salesforce, the #1 AI CRM, today unveiled plans to invest $1 billion in Spain over the next five years,…"

Possible sales relevance: Potentially relevant to prospects evaluating AI CRM investment strategies.

Evidence gap: No explicit mention of IBM as a competitor or partner in the provided sources.

Leadership Intelligence

Verified: Arvind Krishna is Chairman, President and Chief Executive Officer of IBM [S3]

Evidence: "Arvind Krishna Chairman, President and Chief Executive Officer, IBM"

Possible sales relevance: Potentially relevant for sales outreach to IBM's top executive

Meeting Preparation (Suggested Questions)

• A public source states, "IBM and Siemens are collaborating on research to advance IBM's nanostack architecture" What context should we understand before…

• How would Chief Revenue Officer evaluate whether Helps sales teams research prospective customers and prepare for meetings with source-backed… is valuable?

• Which current priorities, constraints, and decision criteria should we understand because they were not verified in public sources?

Sources

[S3] https://newsroom.ibm.com/executive-bios

[S4] https://newsroom.ibm.com/press-releases-research-and-innovation

[S8] https://www.salesforce.com/news/press-releases/2026/10/19/1-billion-ai-leadership-inves…

Step 11. Compare prompts

I compared different prompt structures using the same saved sources. In the final comparison, specialized v3.1 scored 3.57 out of 5 and used 3,409 tokens. Grounded-sales v1.1 scored 3.43 and used 3,806 tokens.

The grounded-sales prompt returned more findings, but more information did not automatically mean better information. This was a small experiment, so the result cannot show that one prompt is always better.

The first baseline prompt also returned the wrong JSON structure. I added a clearer example and category rules before running baseline v1.1. Later prompt versions had similar formatting mistakes, which taught me to test both the prompt wording and the response schema.

Prompt Engineering Comparison

Metric	Specialized v3.1	Grounded-sales v1.1
Manual evaluation score	3.57 / 5	3.43 / 5
Findings retained	2	4
Total tokens	3,409	3,806
API latency	1.403 seconds	1.536 seconds
Exact excerpt validation	100%	100%

Step 12. Test the application

I tested input validation, website retrieval, redirects, blocked pages, uploads, prompt responses, citations, human review, meeting questions, and PDF output.

Final local results:

- 98 automated tests passed
- Ruff passed
- All Python modules compiled
- The sales brief opened as one page

```
src/cap_931/tests/test_document_parser.py::test_malformed_or_mismatched_documents_are_rejected[broken.pdf-%PDF-1.7\nnot a real PDF-malformed or unreadable] PASSED                    [ 59%]
src/cap_931/tests/test_document_parser.py::test_malformed_or_mismatched_documents_are_rejected[renamed.pdf-This is plain text.-valid PDF signature] PASSED                    [ 60%]
src/cap_931/tests/test_document_parser.py::test_malformed_or_mismatched_documents_are_rejected[renamed.txt-%PDF-1.7\n-PDF data] PASSED                    [ 61%]
src/cap_931/tests/test_document_parser.py::test_malformed_or_mismatched_documents_are_rejected[binary.txt-product\x00overview-binary data] PASSED                    [ 62%]
src/cap_931/tests/test_document_parser.py::test_encrypted_pdf_is_rejected PASSED                    [ 63%]
src/cap_931/tests/test_document_parser.py::test_pdf_without_extractable_text_explains_ocr_limit PASSED                    [ 64%]
src/cap_931/tests/test_document_parser.py::test_document_text_is_bounded_for_model_context PASSED                    [ 65%]
src/cap_931/tests/test_experiments.py::test_baseline_schema_context_and_citation_validation PASSED                    [ 66%]
src/cap_931/tests/test_experiments.py::test_baseline_removes_wrong_provenance_and_unknown_ids PASSED                    [ 67%]
src/cap_931/tests/test_experiments.py::test_baseline_rejects_invalid_schema PASSED                    [ 68%]
src/cap_931/tests/test_experiments.py::test_baseline_without_sources_skips_generator PASSED                    [ 69%]
src/cap_931/tests/test_experiments.py::test_baseline_source_union_is_bounded_and_contains_both_provenances PASSED                    [ 70%]
src/cap_931/tests/test_experiments.py::test_mechanical_metrics_do_not_claim_factual_support PASSED                    [ 71%]
src/cap_931/tests/test_experiments.py::test_manual_score_calculation_and_bounds PASSED                    [ 72%]
src/cap_931/tests/test_experiments.py::test_hashing_and_result_serialization_are_reproducible PASSED                    [ 73%]
src/cap_931/tests/test_experiments.py::test_recorded_experiment_artifacts_are_valid_and_hash_matched PASSED                    [ 74%]
src/cap_931/tests/test_experiments.py::test_grounded_variant_separates_facts_relevance_questions_and_gaps PASSED                    [ 75%]
src/cap_931/tests/test_experiments.py::test_grounded_variant_rejects_mismatched_relevance_structure PASSED                    [ 76%]
src/cap_931/tests/test_experiments.py::test_grounded_manual_score_average_and_prompt_versions PASSED                    [ 77%]
src/cap_931/tests/test_experiments.py::test_final_prompt_artifacts_are_valid_and_historical_results_unchanged PASSED                    [ 78%]
src/cap_931/tests/test_llm_client.py::test_groq_request_uses_bounded_low_reasoning_json_mode PASSED                    [ 79%]
src/cap_931/tests/test_sales_tools.py::test_human_review_excludes_findings_but_preserves_gaps PASSED                    [ 80%]
src/cap_931/tests/test_sales_tools.py::test_discovery_questions_use_verified_fact_and_are_labeled_as_questions PASSED                    [ 81%]
src/cap_931/tests/test_sales_tools.py::test_discovery_questions_are_neutral_without_public_findings PASSED                    [ 82%]
src/cap_931/tests/test_step5.py::test_access_denial_is_not_evidence PASSED                    [ 83%]
src/cap_931/tests/test_step5.py::test_company_agent_does_not_use_competitor_only_source PASSED                    [ 84%]
src/cap_931/tests/test_step5.py::test_pdf_one_page_and_source_link PASSED                    [ 85%]
src/cap_931/tests/test_step5.py::test_pdf_text_normalizes_unsupported_hyphens PASSED                    [ 86%]
src/cap_931/tests/test_step5.py::test_pdf_with_reviewed_findings_and_questions_stays_one_page PASSED                    [ 87%]
src/cap_931/tests/test_validators.py::test_valid_public_url PASSED                    [ 88%]
src/cap_931/tests/test_validators.py::test_invalid_urls[] PASSED                    [ 89%]
src/cap_931/tests/test_validators.py::test_invalid_urls[not-a-url] PASSED                    [ 90%]
src/cap_931/tests/test_validators.py::test_invalid_urls[file:///etc/passwd] PASSED                    [ 91%]
src/cap_931/tests/test_validators.py::test_invalid_urls[http://localhost] PASSED                    [ 92%]
src/cap_931/tests/test_validators.py::test_invalid_urls[https://user:pass@example.com] PASSED                    [ 93%]
src/cap_931/tests/test_validators.py::test_invalid_urls[https://example.com:8080] PASSED                    [ 94%]
src/cap_931/tests/test_validators.py::test_rejects_private_dns PASSED                    [ 95%]
src/cap_931/tests/test_validators.py::test_competitor_deduplication PASSED                    [ 96%]
src/cap_931/tests/test_web_research.py::test_extracts_main_content PASSED                    [ 97%]
src/cap_931/tests/test_web_research.py::test_missing_title PASSED                    [ 98%]
src/cap_931/tests/test_web_research.py::test_blocked_html_is_rejected_before_becoming_evidence PASSED                    [100%]

============================================================================== 98 passed in 0.69s ==============================================================================
zara@Zaheeras-MacBook-Pro CAP_931 %
```

Conclusion

I built a working local sales research assistant with a Streamlit interface, three Groq research roles, public website research, source checks, and PDF export.

The most difficult part was not connecting the model. It was finding readable website information and keeping each claim connected to the correct company. Blocked websites also showed me why the app needs clear evidence gaps.

I also had missing-module and function-interface errors while joining the project steps. I fixed them by restoring the package files, checking each module's inputs and outputs, and running the full tests after every integration.

I learned that a source link alone does not prove an AI-generated statement. The prompt comparison also showed me that more findings do not always make a better result.

If I continue this project, I would test more company pairs, add another human reviewer, and improve public-source coverage. I would not treat the current prototype as a production system.

Appendix

Final results

- 98 tests passed
- Four IBM and four Salesforce pages were retrieved in the final displayed run
- The representative sales brief contains one page
- No API key is stored in the publishable files

Final prompt comparison

Approach	Manual score	Findings	Tokens	API latency
Specialized v3.1	3.57 / 5	2	3,409	1.403 s
Grounded-sales v1.1	3.43 / 5	4	3,806	1.536 s

Time management

I worked in stages: interface, validation, Groq connection, website research, document upload, PDF output, prompt testing, and final verification. I do not have a reliable hour-by-hour record, so I did not invent one.

Deployment summary

The app runs locally. A production version would need user login, managed API keys, request limits, private upload rules, monitoring, and stronger network controls. Python URL checks reduce risk, but production would also need restricted outbound network access. The deployment plan explains these items. The project is not currently deployed.

GitHub and references

- GitHub repository
- Project README
- Technical architecture
- Prompt experiments
- Deployment plan
- Final test results
- Sales intelligence brief
- CAP 931 report PDF